Daily Targets Engine – Mathematical Calculation Report

Prepared for Spandan Bole · 28 October 2026 · Calories, protein, carbs, fats, steps and hydration for every age group, sex, objective and lifestyle

1. How the engine works

Every target is computed in a fixed order, and each step feeds the next:

1. Safety gates → 2. BMI → 3. BMR → 4. Climate factor → 5. TDEE → 6. Objective calories → 7. Protein → 8. Fat → 9. Carbs (remainder) → 10. Steps → 11. Hydration → 12. Day-mode split → 13. Meal split.

Conventions. W = weight (kg), H = height (cm), h = H/100 (m), A = age in whole years (auto-updated from date of birth). Energy: protein 4, carbs 4, fat 9 kcal/g. Rounding: calories to 5 kcal, macros to 1 g, steps to 500, water to 10 ml. All numbers are population estimates, so the engine also self-corrects weekly (Section 15).

2. Inputs the app collects

Input	Values	Drives
Sex	Male / Female	BMR constant, fat and calorie floors
Age	From date of birth	BMR equation, protein, steps, water
Height, weight	cm, kg (convert from ft/in, lb)	BMI, BMR, all per-kg targets
Activity level	Sedentary / Light (1–3 d/wk) / Moderate (3–5) / Very Active (6–7)	PAL, protein (Maintain), steps, training days
Objective	Lose Fat / Maintain / Build Muscle	Calories, protein, fat share
Training experience	Beginner (<1 yr) / Intermediate (1–3 yr) / Advanced (>3 yr)	Pace caps, protein, recomposition
Training type	Strength / Cardio / Mixed	Protein tweak, sweat rate, carb guide
Target pace	Gentle / Standard / Fast / Aggressive (fat loss); Slow / Standard / Fast (muscle)	Deficit or surplus
Occupation load	Desk / Light-standing / Active on feet / Heavy manual or outdoor	TDEE add-on, water
Meal frequency	2–6 meals per day	Per-meal split

Input	Values	Drives
Day mode	Uniform Days / Workout-Rest Days	Day-wise targets
Caffeine / alcohol	Yes / No (heavy tea, coffee or alcohol)	Water
Eating out	Rarely / Sometimes / Often	Water (sodium offset)
Climate	Automatic from today's date and location	Water, sweat rate, cold TDEE
Workout minutes per session	Default 30 / 45 / 60 for Light / Moderate / Very Active	Exercise water

3. Safety gates (run first)

- **Age < 13:** no targets; show "talk to a paediatrician or guardian".
- **Age 13–17:** Maintain only (supervised Build Muscle capped at +5% calories). Lose Fat is hidden. Growth needs are not reduced.
- **BMI < 18.5:** Lose Fat is disabled for adults; offer Maintain or Build Muscle. For ages 13–17 apply the same gate when BMI-for-age is below −2 SD (WHO 2007 growth reference).
- **Pregnancy, breastfeeding, diabetes, kidney disease, eating-disorder history:** these formulas do not apply; show a clinician-first notice.
- **Calorie floor (Lose Fat only):** never below max(1,200 kcal female / 1,500 kcal male, 0.9 × BMR). These are conventional safety limits from clinical weight-management practice, not physiological thresholds.
- **Deficit ceiling:** never more than 25% of TDEE.

4. BMI and effective weight

- BMI = W / h^2
- Standard WHO bands: <18.5 underweight; 18.5–24.9 normal; 25–29.9 overweight; ≥30 obese. For ages 13–17 these adult cut-offs do not apply; use BMI-for-age z-scores from the WHO 2007 growth reference (thinness below −2 SD, overweight above +1 SD, obesity above +2 SD).
- Optional Asian-population cut-offs (WHO 2004): increased risk ≥23, high risk ≥27.5. Offer as a setting; it changes the label only, not the formulas.
- **Effective weight** (used for protein, fat floor and water so that high body fat does not inflate targets):
    - If BMI ≤ 27: $W_{eff} = W$
    - If BMI > 27: $W_{ref} = 25 \times h^2$, then $W_{eff} = W_{ref} + 0.25 \times (W - W_{ref})$
- Very muscular users are misclassified by BMI. Provide an override: "I'm muscular / I know my body-fat %", which switches BMR to Katch-McArdle (below). W_{eff} applies to adults only (teens use actual weight). The 0.25 factor follows the adjusted-body-weight convention of clinical nutrition (practice uses about 0.25–0.4); it is a convention, not a measured constant.

5. BMR (basal metabolic rate, kcal/day)

Adults 18+ – Mifflin-St Jeor (1990):

- Male: BMR = 10W + 6.25H − 5A + 5
- Female: BMR = 10W + 6.25H − 5A − 161

Adolescents 13–17 – WHO/FAO/UNU (1985), age band 10–18:

- Male: BMR = 17.5W + 651
- Female: BMR = 12.2W + 746

Optional override when body-fat % (BF) is known – Katch-McArdle: LBM = W × (1 − BF/100); BMR = 370 + 21.6 × LBM.

Accuracy. Predictive equations carry an individual error of roughly ±10% or more. The 1985 WHO/FAO/UNU equations overestimate BMR in many populations, especially in the tropics (Henry 2005), and in a Brazilian urban tropical sample Mifflin-St Jeor overestimated BMR by about 14% (Clin Nutr 2018). Treat BMR as a starting estimate and let the weekly recalibration (Section 15) correct it.

6. Climate factor (from the current date and location)

Every day the app reads today's **feels-like maximum temperature T_a (°C)** for the user's location, which already includes humidity. If no weather data is available, estimate it from the date: choose the hemisphere from the latitude, map the month to a season, then assume T_a = 18 °C (cool season), 26 °C (mild season) or 33 °C (hot or humid season, including monsoon and post-monsoon in the tropics).

T_a (°C)	Water add-on	Exercise sweat multiplier
< 25	+0 ml	× 1.00
25 – 29.9	+250 ml	× 1.00
30 – 34.9	+400 ml	× 1.25
35 – 39.9	+650 ml	× 1.25
≥ 40	+900 ml	× 1.25

The band values are engineering heuristics sized against typical exercise sweat losses of about 0.5–2 L per hour, which rise with heat; they are not published dose-response figures.

- Altitude ≥ 2,500 m: +500 ml extra (heuristic allowance for greater breathing and urine losses).
- Climate does not change the calorie target: the effect of ordinary ambient temperature on energy expenditure is small next to BMR prediction error, so only hydration is adjusted.
- Example: on 28 October 2026 a city such as Mumbai is post-monsoon, with typical highs in the low 30s °C, so the 30–34.9 band (+400 ml) applies. Live data always overrides this fallback.

7. TDEE (total daily energy expenditure)

TDEE = BMR × (activity factor + occupation add-on)

Activity level	PAL
Sedentary (work, little exercise)	1.200
Light (1–3 days/week)	1.375
Moderate (3–5 days/week)	1.550
Very Active (6–7 days/week)	1.725

In these tables “PAL” means this activity factor. The multipliers are the conventional Harris-Benedict/McArdle values paired with BMR equations; they are not the FAO/WHO/UNU PAL categories, which are defined against measured BMR and span about 1.4–2.4. Treat the result as a starting estimate.

Occupation load	Add-on to PAL
Desk	+0.00
Light-standing (teacher, retail)	+0.05
Active on feet (nurse, courier)	+0.10
Heavy manual / outdoor	+0.15

No standard table exists for occupation add-ons; these are engineering estimates, kept small to limit double-counting with the activity tier.

- Teens 13–17: add +0.10 to the activity factor as a heuristic allowance for growth and higher activity (adolescent energy needs per kg are higher than adults'); weekly recalibration corrects individual error.
- Cap the total (activity factor + occupation add-on) at 2.0.
- Training type does **not** change TDEE, because the activity tier already represents training volume. Counting it again would double-count.

8. Objective, pace and training age → daily calories (C)

8.1 Maintain

C = TDEE.

8.2 Lose Fat

Pace is a percentage of body weight lost per week:

Pace	r (fraction of W per week)
Gentle	0.0025
Standard	0.0050
Fast	0.0075

Pace	r (fraction of W per week)
Aggressive	0.0100

The chosen pace is capped (take the lowest cap that applies): Beginner ≤ 1.0%, Intermediate ≤ 0.75%, Advanced ≤ 0.5%; BMI < 22 ≤ 0.5%; BMI 22–25 ≤ 0.75%; age ≥ 65 ≤ 0.5%.

- Deficit = min(1,100 × W × r, 0.25 × TDEE), using 7,700 kcal per kg of fat ÷ 7 days = 1,100 kcal per kg per week
- C = max(TDEE − Deficit, calorie floor from Section 3)

The 7,700 kcal per kg figure is a short-term approximation: the real energy density of weight change depends on the fat-to-lean mix, and metabolic adaptation shrinks a deficit over time (Hall 2011), which the weekly recalibration absorbs. The pace caps are conservative engineering limits within the commonly advised 0.5–1% of body weight per week.

8.3 Build Muscle

Pace	Surplus s
Slow	5%
Standard	10%
Fast	15%

Caps by training age: Beginner ≤ 15%, Intermediate ≤ 12%, Advanced ≤ 8%. If BMI ≥ 27 the cap is 5% (lean gain). Teens ≤ 5%.

- C = TDEE × (1 + s_final)

The surplus caps follow off-season guidance of roughly 5–20% above maintenance, higher for beginners and lower for advanced lifters (Iraki 2019); the engine stays at or below the middle of that range.

8.4 Why training age matters

- **Beginners** gain muscle and lose fat at the same time (body recomposition), so they can run a **standard macro split even in a deficit**: protein 1.6 g/kg, fat 25%, carbs fill the rest.
- **Advanced** trainees recompose far less readily; recomposition is mainly seen in beginners, returning lifters and people with more body fat (Barakat 2020). They get a **protein bias** (2.4 g/kg, 50% more than beginners), a slower pace cap (0.5%/week) and a lower fat share (22%) so carbs still fuel training.

9. Protein (P, grams)

P = (base g/kg + modifiers) × W_eff

Objective	Base g/kg
Maintain – Sedentary / Light / Moderate / Very Active	0.9 / 1.1 / 1.3 / 1.5

Objective	Base g/kg
Lose Fat – Beginner / Intermediate / Advanced	1.6 / 2.0 / 2.4
Build Muscle – Beginner / Intermediate / Advanced	1.6 / 1.8 / 2.0

Basis: the RDA of 0.8 g/kg is a minimum for sedentary adults; ISSN (Jäger 2017) puts exercising adults at 1.4–2.0 g/kg; Morton 2018 finds muscle-gain benefit levelling off near 1.6 g/kg (upper confidence bound 2.2). In a deficit, lean trainees need more: Helms et al. recommend 2.3–3.1 g per kg of lean body mass, roughly 1.8–2.7 g/kg of total weight at typical body-fat levels, hence 2.4 for Advanced. Beginners can recompose on 1.6 g/kg.

Modifiers:

- Cardio-only training: −0.1 g/kg (Maintain and Build only; floors 0.8 and 1.4 g/kg).
- Age ≥ 60: +0.3 g/kg, to slow age-related muscle loss (PROT-AGE recommends 1.0–1.2 g/kg for healthy older adults, more with exercise or illness).
- Teens 13–17: fixed 0.95 g/kg (Sedentary or Light) or 1.2 g/kg (Moderate or Very Active), ignoring the table (the RDA is 0.85 g/kg for ages 14–18 and 0.95 g/kg for ages 9–13; the higher value covers active teens).
- Cap: $P \le \min(2.5 \times W_{eff}, 0.35 \times C / 4)$. Floor: $0.8 \times W_{eff}$.
- Sex does not change the per-kg value; it is already reflected in W.

10. Fat, carbs, fibre and salt

Fat (F, grams):

- Share of calories: Lose Fat 25% (Advanced 22%), Maintain 28%, Build Muscle 25%; teens and age ≥ 60 use 30%.
- $F = \text{share} \times C / 9$
- Floor: $\max(0.5 \text{ g/kg male or } 0.6 \text{ g/kg female} \times W_{eff}, 20\% \times C / 9)$. Ceiling: $30\% \times C / 9$ for adults (WHO 2023) and $35\% \times C / 9$ for teens (US/Canada AMDR).

Basis: WHO (2023) advises adults to keep total fat at or below 30% of energy, and the US/Canada AMDR is 20–35% (25–35% for ages 4–18); every share above sits inside both. The g/kg floors are conventional practice values, not RDAs. WHO (2023) also considers carbohydrate intakes of 40–70% of energy compatible with health, with quality (whole grains, vegetables, fruit, pulses) mattering most; the example plans fall at 47–57%.

Carbohydrate (K, grams) – the remainder:

- $K = (C - 4P - 9F) / 4$
- Minimum 130 g/day (the US/Canada RDA for adults, set from the brain's glucose use). If K falls below it, reduce fat to its floor first; if still short, flag the plan as too aggressive.
- Advisory fuel range for training days (g/kg): Strength 3–5, Mixed 4–6, Cardio 5–8, adapted from the ACSM/AND/DC 2016 bands (3–5 g/kg light, 5–7 moderate at about 1 h/day, 6–10 at 1–3 h/day). In a deficit it is acceptable to sit below the range.

Companion guides:

- Fibre = 14 g per 1,000 kcal (IOM adequate intake); for adults use at least 25 g/day, the WHO (2023) minimum.
- Sodium < 2,000 mg/day and free sugars < 10% of calories (both WHO).

11. Daily steps

Steps = round_to_500(clamp(S_tier + A_objective))

Activity level	S_tier		Objective	A_objective
Sedentary	5,000		Lose Fat	+1,500
Light	7,000		Maintain	0
Moderate	8,500		Build Muscle	−500
Very Active	10,000			

Basis: the tier values sit at Tudor-Locke's adult bands (under 5,000 sedentary; 5,000–7,499 low active; 7,500–9,999 somewhat active; 10,000 and above active). The objective adjustments (+1,500 and −500) are engineering choices: more daily movement raises expenditure during fat loss, slightly less conserves energy during a surplus.

- Age 18–59: clamp to 4,000–12,000. Age ≥ 60: clamp to 4,000–8,000 (health benefit plateaus around 6,000–8,000). Teens: minimum 10,000 (about 10,000–11,700 steps/day corresponds to 60 minutes of moderate-to-vigorous activity in adolescents; Tudor-Locke 2011).
- Distance (km) = Steps × stride / 1,000, with stride = 0.415 × H cm for men and 0.413 × H cm for women, divided by 100 to give metres.
- Calories burned by walking ≈ Steps × W × 0.0004 (a net walking cost of about 0.5 kcal per kg per km). This is shown for information only and is **not** added to calories, because PAL already includes it.

12. Hydration (drinking fluids, ml/day)

Water = base + exercise + climate + occupation + caffeine/alcohol + eating out, then clamped.

Base = k × W_eff

Age	k (ml/kg)
13–17	38
18–64	32
65+	30

Basis: adequate intakes of total water are 2.0 L (women) and 2.5 L (men) per day (EFSA 2010), or 2.7 L and 3.7 L (US NASEM 2005), with roughly 20% coming from food. The ml/kg values are this engine's weight-scaling of those figures, not published per-kg recommendations, and the target counts drinks only (tea, coffee and soups count). Needs vary widely between people, so thirst and pale urine remain the practical check.

Add-on	Rule
Exercise (workout days only)	minutes × rate × sweat multiplier. Rate: Strength 8, Mixed 10, Cardio 13 ml/min; multiplier from Section 6
Climate	Band in Section 6 (+ altitude if applicable)
Occupation	Desk +0; Light-standing +150; Active on feet +300; Heavy manual/outdoor +500 ml
Caffeine / alcohol = Yes	+300 ml (precautionary buffer for heavy caffeine or alcohol use; see the evidence note below)
Eating out	Rarely +0; Sometimes +150; Often +350 ml, a precaution for high-sodium restaurant meals

Evidence note: the exercise rates (480–780 ml per hour) sit inside typical exercise sweat losses, and ACSM advises replacing fluid so that body-mass loss stays under about 2% (Sawka 2007). Climate, occupation and eating-out values are engineering heuristics. For caffeine, moderate habitual coffee intake (about 4 cups a day) did not differ from water on hydration markers in a controlled trial (Killer 2014), and caffeinated drinks still count as fluid; alcohol does raise urine output, mainly spirits and strong drinks. The +300 ml is therefore a precaution the user can switch off, not a measured requirement. Sodium itself is best managed through the < 2,000 mg/day goal rather than extra water.

- Floor: 1,500 ml adults, 1,200 ml teens. Ceiling: 5,000 ml (and never more than roughly 0.8–1 litre per hour, near the kidneys' maximum water excretion rate, to avoid hyponatraemia).
- Reminder spacing = Water ÷ 16 waking hours; glasses = Water ÷ 250.
- Not valid for people on fluid restriction (heart or kidney conditions).

13. Uniform Days vs Workout/Rest Days

Uniform Days: every day equals the target C, protein, fat, carbs, and water = weekly average of the workout and rest values.

Workout/Rest Days: the weekly calorie total stays exactly 7 × C.

- Training days per week n_w: Sedentary 0, Light 2, Moderate 4, Very Active 6 (user can override). Rest days $n_r = 7 - n_w$. If $n_w = 0$ or $n_r = 0$, use Uniform.
- Swing ϕ_{exp}: Beginner 0.05, Intermediate 0.08, Advanced 0.10.
- **$\phi = \min(\phi_{exp}, 0.15 \times n_r / n_w)$**, so a rest day never drops below 85% of C.
- Workout-day calories $C_w = C \times (1 + \phi)$
- Rest-day calories $C_r = C \times (1 - \phi \times n_w / n_r)$
- Check: $n_w \times C_w + n_r \times C_r = 7C$.
- Protein and fat stay constant on both days. Carbs absorb the change: $K_w = K + (C_w - C)/4$ and $K_r = K - (C - C_r)/4$. If $K_r < 130$ g, lower ϕ until it holds.
- Water uses the workout-day value (with exercise) and the rest-day value (without).
- Steps are the same on both days.

Evidence note: weekly energy balance, not its day-to-day distribution, drives weight change, and there is little evidence that calorie or carbohydrate cycling beats uniform intake for body composition. The split is offered for preference, training fuelling and adherence; the ϕ values are engineering choices.

14. Meal frequency split

Meals n ∈ {2…6}. Three are main meals (weight 1.0); extra meals are snacks (weight 0.5). With n ≤ 3 all meals are mains.

- Share of meal i = w_i / Σw
- Meal calories = C × share; meal protein = P / n (equal per meal, snacks included).
- Check: protein per meal should sit between about 0.25 and 0.55 g/kg W_eff (Schoenfeld & Aragon 2018). Above that, suggest adding a meal; below it, suggest merging meals.
- Meal frequency itself has little effect on fat loss when calories and protein are matched (Schoenfeld 2015); it is for adherence and protein spacing.

15. Weekly recalibration

Every 14 days compare the average weekly weight change with the plan (rates in kg/week, gain positive):

C_new = C_old + (planned_rate − actual_rate) × 1,100, limited to ±150 kcal per update. Recompute macros afterwards with the same rules. Ignore the first week after a change in sodium or carbs, since water weight is noisy.

16. Age and sex summary

Group	BMR	PAL tweak	Objectives	Protein	Fat	Water	Steps
Teen 13–17, M/F	WHO/FAO equation	+0.10	Maintain (Build ≤ +5%)	0.95–1.2 g/kg	30%	38 ml/kg	≥ 10,000
Adult 18–59, male	Mifflin, +5	none	All	per Section 9	20–30%, floor 0.5 g/kg	32 ml/kg	≤ 12,000
Adult 18–59, female	Mifflin, −161	none	All	per Section 9	20–30%, floor 0.6 g/kg	32 ml/kg	≤ 12,000
60–64, M/F	Mifflin	none	All, Lose Fat paced ≤ 0.5% if 65+	+0.3 g/kg	30%	32 ml/kg	≤ 8,000
65+, M/F	Mifflin	none	All, pace ≤ 0.5%	+0.3 g/kg	30%	30 ml/kg	≤ 8,000

Sex differences enter through BMR, the calorie floor (1,200 vs 1,500 kcal), the fat floor and body size; per-kg protein and carbs are the same.

17. Worked examples

Case A – Male, 28 y, 178 cm, 80 kg, Moderate, Lose Fat

Intermediate, Mixed training (45 min), Standard pace, 4 meals, desk job, caffeine = Yes, eating out = Sometimes, T_a = 32 °C.

- BMI = 80 / 1.78^2 = 25.3 (W_{eff} = 80)
- BMR = 800 + 1,112.5 − 140 + 5 = 1,777.5 kcal
- TDEE = 1,777.5 × 1.55 = 2,755 kcal
- Deficit = 1,100 × 80 × 0.005 = 440 kcal (below the 689 kcal ceiling) → **C = 2,315 kcal**
- Protein = 2.0 × 80 = **160 g** (640 kcal, 28%)
- Fat = 25% × 2,315 / 9 = **64 g** (576 kcal, 25%)
- Carbs = (2,315 − 640 − 576) / 4 = **275 g** (1,100 kcal, 47%)
- Fibre ≈ 32 g
- Steps = 8,500 + 1,500 = **10,000** (≈ 7.4 km)
- Water, rest day = 2,560 + 400 (climate) + 300 (caffeine) + 150 (eating out) = **3,410 ml**
- Water, workout day = 3,410 + 10 × 45 × 1.25 = **3,970 ml**
- Uniform water = (4 × 3,972.5 + 3 × 3,410) / 7 = **3,730 ml**
- Workout/Rest split: n_w = 4, n_r = 3, ϕ = min(0.08, 0.15 × 3/4 = 0.1125) = 0.08
    - Workout day: C_w = **2,500 kcal**, carbs 321 g
    - Rest day: C_r = **2,070 kcal**, carbs 213 g
- Meals (4): weights 1, 1, 1, 0.5 → 660 kcal per main and 330 kcal per snack; protein 40 g per meal (0.50 g/kg, inside the 0.25–0.55 range).

Case B – Female, 24 y, 165 cm, 58 kg, Light, Build Muscle

Beginner, Strength (45 min), Standard pace, desk job, T_a = 18 °C, caffeine = No, eating out = Rarely.

- BMI = 58 / 1.65^2 = 21.3
- BMR = 580 + 1,031.25 − 120 − 161 = 1,330 kcal
- TDEE = 1,330 × 1.375 = 1,829 kcal
- Surplus 10% (cap 15%) → **C = 2,010 kcal**
- Protein = 1.6 × 58 = **93 g** (18%)
- Fat = 25% × 2,012 / 9 = **56 g** (floor 35 g)
- Carbs = (2,012 − 372 − 504) / 4 = **284 g** (4.9 g/kg, inside the strength range)
- Fibre ≈ 28 g; Steps = 7,000 − 500 = **6,500**
- Water, rest day = 32 × 58 = **1,860 ml**; workout day = 1,856 + 8 × 45 = **2,220 ml**; uniform = **1,960 ml**
- Workout/Rest split: n_w = 2, n_r = 5, ϕ = min(0.05, 0.375) = 0.05 → C_w = **2,115 kcal** (carbs 309 g), C_r = **1,970 kcal** (carbs 274 g)

Case C – Female, 67 y, 158 cm, 70 kg, Sedentary, Maintain

No training, retired, T_a = 32 °C, caffeine = No, eating out = Rarely.

- BMI = 70 / 1.58² = 28.0 → W_ref = 62.4, W_eff = 62.4 + 0.25 × 7.6 = 64.3 kg
- BMR = 700 + 987.5 − 335 − 161 = 1,191.5 kcal
- TDEE = 1,191.5 × 1.2 = 1,430 kcal → **C = 1,430 kcal**
- Protein = (0.9 + 0.3) × 64.3 = **77 g** (22%)
- Fat = 30% (the age ≥ 60 share) × 1,430 / 9 = 48 g; floor 0.6 × 64.3 = 39 g, so **48 g** (30%)
- Carbs = (1,430 − 308 − 432) / 4 = **173 g** (48%, above the 130 g floor)
- Fibre = 25 g (WHO adult minimum; 14 g per 1,000 kcal would give only 20 g); Steps = **5,000** (stretch goal 6,000)
- Water = 30 × 64.3 + 400 (climate) = **2,330 ml**, with scheduled reminders because thirst weakens with age
- Sedentary → n_w = 0, so Uniform Days only.

18. Evidence base and limits

- BMR: Mifflin et al. 1990; FAO/WHO/UNU 1985 (adolescents); Katch-McArdle; Henry 2005 and Clin Nutr 2018 (overestimation in tropical populations); WHO 2007 growth reference (BMI-for-age).
- Protein: ISSN position stand (Jäger 2017); Morton et al. 2018 meta-analysis (1.6 g/kg plateau, up to 2.2); Helms et al. 2014 (2.3–3.1 g per kg lean mass for lean athletes in a deficit); PROT-AGE 2013 (older adults); Barakat 2020 (recomposition); Iraki 2019 (off-season surplus ranges); Schoenfeld 2015 and Schoenfeld & Aragon 2018 (meal frequency, protein per meal).
- Energy density 7,700 kcal/kg is a short-term approximation (Hall 2011); this is why pace caps and the 14-day recalibration exist.
- Fat, carbohydrate and fibre: WHO 2023 guidelines (total fat ≤ 30% of energy for adults; carbohydrate 40–70% of energy; fibre ≥ 25 g/day); US/Canada DRI 2005 (AMDR fat 20–35% and carbohydrate 45–65%, carbohydrate RDA 130 g/day, fibre 14 g per 1,000 kcal); WHO sodium (< 2 g/day) and free-sugar (< 10% of energy) guidelines.
- Steps: Tudor-Locke 2011 (adult activity bands, youth guidance); Paluch et al. 2022 (mortality benefit plateau about 6,000–8,000 steps if ≥ 60, 8,000–10,000 if younger).
- Hydration: EFSA 2010 and NASEM 2005 adequate intakes; ACSM position stand on exercise and fluid replacement (Sawka 2007); Killer 2014 (coffee vs water). The per-kg scaling and the climate, occupation, caffeine/alcohol and eating-out add-ons are engineering heuristics, kept conservative.
- These are general-population estimates and starting points, not medical advice; individual error is large, which is why the weekly recalibration exists. Anyone with a medical condition, pregnancy, an eating-disorder history or a minor user should be directed to a qualified professional.